Aakash Tripathi
Bachelor's in Computer Science, George Mason University, Fairfax VA, May 2026 Graduate
Professional Certificate in AI and Machine Learning via Michigan Engineering Professional Education and IBM (In Progress)
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Technical Skills

- Languages: Java, Python, SQL, JavaScript, C, C++
- Frameworks & Tools: Flask, React, Git, GitHub, Bitbucket, JIRA, VS Code, Android Studio
- Cloud & DevOps: AWS EC2, S3, AWS CLI, Jenkins, Ansible
- Databases: SQLite, Firebase Realtime Database, ChromaDB, SQLAlchemy, PostgreSQL
- Certifications: AWS Certified Cloud Practitioner
- Relevant Coursework: Intro to AI, Android App Dev, Data Structures and Algorithms, Agentic AI Systems – DeepLearning.AI (Andrew Ng)

Experience

Leidos — Fairfax, VA
Undergraduate ETL Developer (Industry-Sponsored Project) — August 2025 – May 2026

- Designed and developed an end-to-end ETL pipeline to transform IFC building models into VentSim-ready simulation networks, enabling airflow analysis for engineering applications.
- Parsed IFC data using IfcOpenShell to extract ducts, walls, and ports and construct topology graphs of connected systems.
- Architected a graph-based data model to represent duct connectivity, enabling scalable transformation into simulation-ready network structures
- Engineered transformation logic to generate VentSim outputs, including geometry normalization and airway dimension generation.
- Designed a solution to handle missing fan data in IFC models by integrating external mappings into simulation workflows.

Freddie Mac — McLean, VA
Software Engineer Intern — May 2025 – August 2025 | May 2024 – August 2024

- Redesigned and optimized Python-based data ingestion pipelines, improving execution time by 20% and increasing system reliability.
- Built automated unit tests in Python (pytest) to validate JSON-to-Excel data pipelines, improving data accuracy and reducing manual verification effort.
- Utilized AWS EC2 instances to test and push updates to Bitbucket repositories, ensuring version-controlled CI/CD integration.
- Developed Ansible playbooks to automate dataset queries, effectively identifying and reporting critical issues for Problem Management, improving resolution timelines.
- Proposed a $77K automation initiative by translating complex ServiceNow ticket dependencies into structured workflows using existing Freddie Mac systems; received director approval and executive-level interest.

Projects

Pep.AI – AI Football Scouting Platform (https://github.com/Acash2018/Pep.AI)

- Implemented a ChromaDB-powered RAG system over 14+ football intelligence documents, including tactical systems, player roles, scouting reports, and tactical analysis.
- Designed a 0-100 tactical fit scoring system using role suitability, formation fit, tactical strengths, risk factors, and retrieved football knowledge.
- Built a full-stack AI scouting platform that analyzes football players across tactical systems using a 3-agent LangGraph workflow: Stats Agent, Tactical Fit Agent, and Report Writer Agent.
- Ingested 100+ public StatsBomb player profiles from Bundesliga match data and transformed event-level data into scouting profiles with inferred strengths, weaknesses, tactical roles, and fit scores.